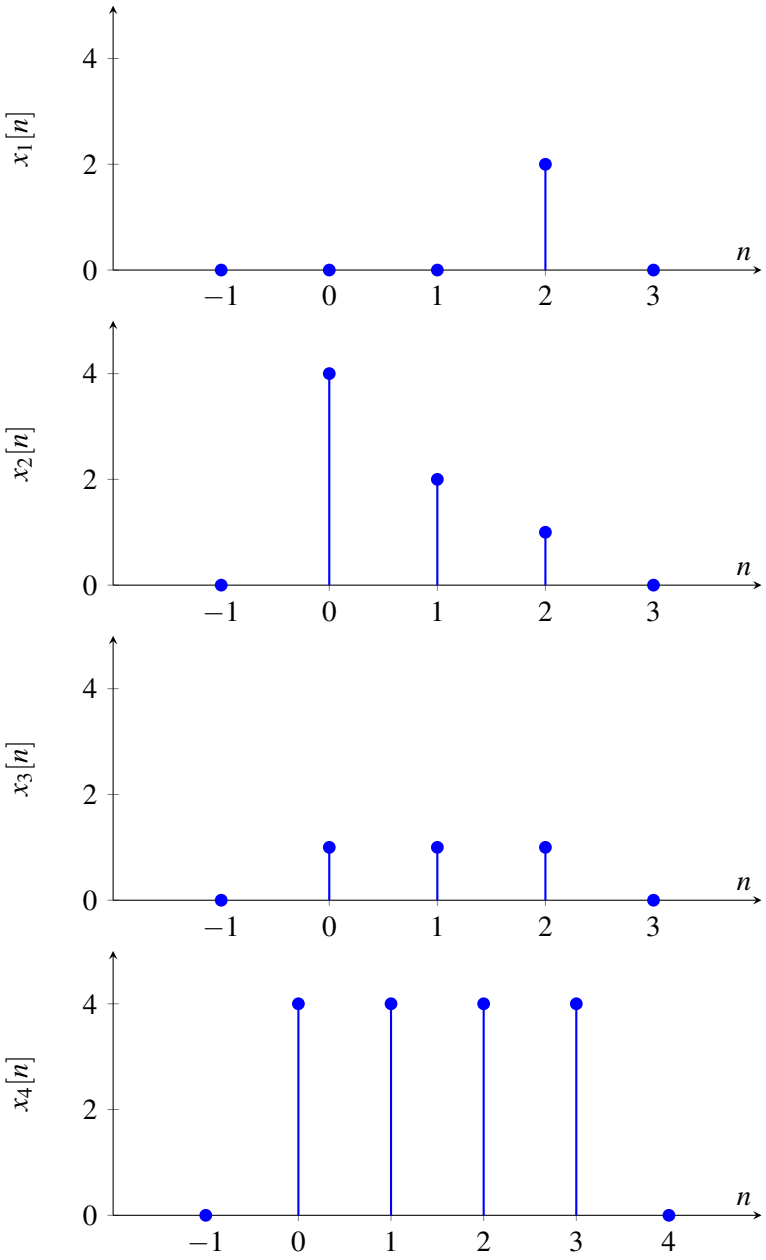
# EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 10A

**1. DT Convolution Practice [Inspired by EE 120]**

Consider the four discrete-time signals below.




Carefully sketch and label plots of the following convoluted signals.

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(a) $(x_1 * x_2)[n]$

(b) $(x_3 * x_3)[n]$

(c) $(x_3 * x_4)[n]$

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## 2. Rolling Dice with Convolution

Let $X$ and $Y$ be the values of two independent rolls of a fair six-sided die. The probability mass function (PMF) of $X$ assigns a probability to each integer:

$$p_X[n] = \Pr(X = n) = \begin{cases} \frac{1}{6}, & n \in \{1,2,3,4,5,6\}, \\ 0, & \text{otherwise.} \end{cases}$$

$Y$ has the same PMF, $p_Y[n] = p_X[n]$. Since the PMF is defined for every integer $n$, we can treat it as a discrete-time signal.

(a) Sketch $p_X[n]$ as a discrete-time signal. Write $p_X[n]$ as a sum of shifted impulses $\delta[n-k]$.

(b) Let $Z = X + Y$. The event $\{Z = n\}$ happens exactly when $X = k$ and $Y = n - k$ for some integer $k$, and these events are disjoint for different values of $k$. Use this fact and the independence of $X$ and $Y$ to show that

$$p_Z[n] = \Pr(Z = n) = \sum_{k=-\infty}^{\infty} p_X[k]\, p_Y[n-k] = (p_X * p_Y)[n].$$

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(c) Compute $p_Z[n]$ for all integers $n$ by evaluating the convolution. Sketch $p_X[k]$ and $p_Y[n-k]$ as functions of $k$ for $n = 4$, and count the values of $k$ where both are nonzero. Then sketch $p_Z[n]$.

(d) Use the flip-and-shift picture to explain why 7 is the most likely sum. Why is $p_Z[n]$ nonzero for exactly 11 values of $n$? As a check, verify that $\sum_n p_Z[n] = 1$.